

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – EXPERIMENTS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO3: Articulation (BL4, BL5)	Assessment:	Assessment Tool 1 – Experiments
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication		
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Section / Batch:		Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Write the algorithm and execute the code for the given experiment.
- Follow a well-structured, systematic approach to design and implement an optimal solution.
- Provide insightful analysis with accurate validation, supported by clear documentation including diagrams, code, and results.

Question Type	Focus Area	Questions
Experiments	Design and Code for Divide and Conquer Algorithms: Merge Sort, Quick Sort, Binary Search, and Matrix Multiplication	Q1

SECTION A – Experiments

Code of Conduct

I _____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Pseudocode:

(03-AT-1)
EXPERIMENTS

1. $t, p = \text{input}(), \text{input}()$
for i in $\text{range}(\text{len}(t) - \text{len}(p) + 1)$:
if $t[i:i + \text{len}(p)] = p$: $\text{print}(i)$;]
break.
2. $t = \text{input}()$
 $p = \text{input}()$
 $n = \text{len}(t)$
 $m = \text{len}(p)$
 $\text{Print}(\text{find}(p))$
3. $m = [\text{list}(\text{map}(\text{int}, \text{input().split()}))]$
for i in $\text{range}(3)$:
 $k = \text{int}(\text{input}())$
for i in $\text{range}(3)$:
for j in $\text{range}(3)$:
if $m[i][j] == k$: $\text{print}(i+1, j+1)$
4. $t = \text{input}()$
 $p = \text{input}()$
for i in $\text{range}(\text{len}(t) - \text{len}(p) + 1)$:
 $\text{Print}(i, \text{"match"}, \text{if } t[i:i + \text{len}(p)] == p$
else ("No match")

```

5. a = list(map(int, input().split()));
   k = int(input())
   print([i for i in range(len(a)) if a[i] == k])

```

```

6. a = list(map(int, input().split(", ")));
   k = int(input())
   print(a → index k+1)

```

```

7. t = input();
   p = input();
   print([i for i in range(len(t) - len(p) + 1)
          if t[i:i + len(p)] == p])

```

```

8. a = list(map(int, input().split(", ")));
   t = int(input())
   print("Not Found");
   print("Comparisons: ", len(a))

```

```

9. a = list(map(int, input().split(", ")));
   for k in map(int, input().split(", ")):
       print(a.index(k+1))
   if k in a else -1

```

```
10. t = input()
    p = input()
    Print("Matches:", t.count(p))
```

```
11. a = list(map(int, input().split()));
```

```
    k = int(input())
```

```
    a.append(k);
```

```
    Print(a.index(x)+1)
```

```
12. a = list(map(int, input().split(",")));
```

```
    k = int(input())
```

```
    Print(a.index(k)+1)
```

```
13. t = input(),
```

```
    p = input().
```

```
    Print([i+1 for i in range(len(t)-len(p)+1)
```

```
        if t[i:i+len(p)] == p])
```

```
14. t = input.lower();
```

```
    p = input().lower();
```

```
    Print(t.find(p)+1)
```

```
15. a = input().split(",");
```

```
    s = input()
```

```
print(a.index(s)+1)
```

```
16. t = input()
    p = input()
    print([i+1 for i in range((len(t)-len(p)+1)
    if t[i:i+len(p)] == p])])
```

```
17. a = list(map(int, input().split()));
    k = int(input());
    print(a.count(k))
```

```
18. t = input();
    p = input()
    for i in range(len(t)-len(p)+1);
    if t[i:i+len(p)] == p:
        print(i+1)
    print("Best: o(n) worst: o(n*m)")
```

```
19. a = list(map(int, input().split()));
    k = int(input())
    print(a.index(k)+1)
```

```
20. t, p1 = input()
```

```
    p2 = input()
```

```
    print(t.count(p1), t.count(p2))
```

RUBRICS FOR CALCULATION

Experiments					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding ; unable to integrate concepts
Experimental Design	30%	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20%	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15%	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10%	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%				
Experimental Design	30%				
Use of Tools	20%				
Analysis of Results	15%				
Documentation	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____